

Mia Francis Bonini

mbonini@umich.edu

(916) 365-7411

[linkedin.com/in/mia-bonini](https://www.linkedin.com/in/mia-bonini)

SUMMARY

PhD biomedical engineer specializing in computational modeling of cardiovascular physiology and interventions. Experienced in developing patient-specific multi-scale CFD and fluid–structure interaction models integrating clinical imaging and hemodynamic data to evaluate valve disease, LVAD support, and aortic interventions. Proven ability to translate in-silico simulations into clinically relevant insights through collaboration with engineers, cardiac surgeons, and radiologists. Strong track record of building validated modeling pipelines to study device performance, procedural outcomes, and cardiovascular mechanics.

EDUCATION

Ph.D. in Biomedical Engineering University of Michigan, Ann Arbor, MI <i>Dissertation Title: Patient-Specific Computational Modeling of Cardiovascular Interventions and Pathologies</i>	Nov. 2025
Master of Science in Biomedical Engineering University of Michigan, Ann Arbor, MI	Apr. 2022
Bachelor of Science in Mechanical Engineering University of Portland, Portland, OR	May 2020
• University of Applied Sciences Upper Austria	Jan. - Apr. 2018

RESEARCH EXPERIENCE

University of Michigan, Cardio Biomechanics Lab Research Staff Advisor: David Nordsletten	Ann Arbor, MI Nov. 2025 – current
• Lead development of an image-based fluid–structure interaction (FSI) modeling framework for patient-specific simulation of aortic disease and cardiovascular device interventions, enabling in-silico evaluation of treatment strategies and hemodynamic risk factors.	
Graduate Student Research Assistant Advisor: David Nordsletten	Aug. 2020 – Nov. 2025
• Designed and validated a patient-specific 3D–0D CFD modeling pipeline integrating CT-derived geometries with lumped circulatory models, achieving <10% error versus patient hemodynamic measurements. • Conducted patient-specific CFD studies of aortic valve replacement and valve-in-valve procedures, demonstrating that annular enlargement reduces transvalvular gradients by 92% without increasing thrombosis risk. • Simulated valve repair strategies in patient-specific LVAD heart models using incompressible Navier–Stokes and scalar transport simulations to evaluate hemodynamics and thrombosis risk. • Performed in-silico experiments to optimize fenestration size and placement for a novel aortic dissection device, informing surgical design decisions and supporting a \$415k funded device development effort. • Applied the 3D–0D framework in mitral regurgitation patients to systematically vary model parameters, demonstrating that conventional clinical metrics can underestimate right ventricular burden. • Organized and led meetings with engineers, cardiac surgeons, radiologists, and students to ensure project completion, leading to 5 first-author articles, 7 co-authored articles, 2 book chapters, and 7+ presentations at national and international conferences.	
Wake-Forest University, Department of Biomedical Engineering Undergraduate Researcher	Winston-Salem, NC Jun – Aug. 2019
• Developed an automated imaging analysis pipeline to quantify long-term spaceflight–induced changes in spinal muscle morphology and composition, reducing manual processing time and improving reproducibility.	

- Utilized OsiriX to straighten cervical spine MRI datasets and Materialise Mimics to perform semi-automated segmentation of paraspinal muscles, enabling precise quantification of cross-sectional area and intramuscular fat infiltration.
- Integrated subject-specific imaging and morphometric data into the finite element Global Human Body Models Consortium (GHBMC) framework to inform biomechanical simulations of spinal loading and musculoskeletal adaptation.

WORK EXPERIENCE

Intel Corporation

Folsom, CA

Engineering Intern – Solid-State Drive & Thermal Solutions

Jun. – Aug. 2018

- Prepared and experimentally evaluated solid-state drives (SSDs) with integrated thermal management solutions in wind tunnel and still-air chamber environments, analyzing temperature profiles to optimize device thermal performance and reliability.
- Designed and executed four-point bending experiments on SSD assemblies using strain gauges and data acquisition systems, quantifying stress-strain response and deflection to improve mechanical robustness.
- Standardized laboratory testing protocols and documentation to reduce operator variability, improve reproducibility, and enhance workflow reliability across mechanical and thermal validation processes.

SKILLS & INTERESTS

Technical: finite element analysis (FEA), computational fluid dynamics (CFD), fluid-structure interaction (FSI), hemodynamic modeling of medical devices, in-silico valve deployment, in-silico experimentation, medical image segmentation, simulations, statistics, scientific writing and communication

Image Processing: 3D Slicer, Materialize Mimics, ITK Snap, Eidolon, OsiriX

Computer: MATLAB, Python, Linux Bash, Ansys, COMSOL, Abaqus

Personal: skiing, volleyball, reading, card games, hiking

SELECTED PUBLICATIONS

1. **Bonini M**, et al. A Monolithic Patient-Specific 3D–0D Model for In-Silico Investigation of Hemodynamics in Patients with Left Ventricle Assist Devices (LVAD). *Biomechanics and Modeling in Mechanobiology*. 2026. (Accepted).
2. **Bonini M**, et al. Image-Based, Whole-System Hemodynamic Modeling of Mitral Regurgitation and Its Impact on the Right Ventricular Function. *Frontiers in Cardiovascular Medicine*; 2026. doi:12:1673443. doi:10.3389/fcvm.2025.1673443
3. **Bonini M**, Hirschvogel M, et al. Patient-Specific Hemodynamic Modeling for Patients with Left Ventricle Assist Device. In: *Functional Imaging and Modeling of the Heart*. Lecture Notes in Computer Science; 2025. doi:10.1007/978-3-031-94562-5_29
4. **Bonini M**, Sanjay S, et al. Modeling the Hemodynamic Impact of Y-incision Aortic Annular Enlargements on Aortic Valve Replacement and Valve-in-Valve Procedures. *Journal of Cardiovascular and Translational Research*. 2025;18:876–887. doi:10.1007/s12265-025-10634-x
5. **Bonini M**, et al. Modeling the Hemodynamic Impact of Mitral and Aortic Valve Repair in Patients with Left Ventricular Assist Devices. *Computers in Biology and Medicine*. 2026. (In submission).
6. Sayut JP, Jilberto J, **Bonini M**, Hirschvogel M, Zhang W, Nordsletten DA. In Silico Evaluation of Cardiac Tissue-Engineered Patch Interventions. *Computers in Biology and Medicine*. 2025;198(Pt B):111249. doi:10.1016/j.combiomed.2025.111249
7. Hirschvogel M, **Bonini M**, Balmus M, Nordsletten D. Effective Block Preconditioners for Fluid Dynamics Coupled to Reduced Models of a Non-Local Nature. *Computer Methods in Applied Mechanics and Engineering*. 2025;435:117541. doi:10.1016/j.cma.2024.117541
8. Hirschvogel M, Balmus M, **Bonini M**, Nordsletten D. Fluid-reduced-solid interaction (FrSI): Physics- and projection-based model reduction for cardiovascular applications. *Journal of Computational Physics*. 2024;506:112921. doi:10.1016/j.jcp.2024.112921

BOOK CHAPTERS

1. **Bonini M**, Jilberto J, Hirschvogel M, Nordsletten D. (2027) Biomechanics of the Heart: From Model to Patient. In Bovendeerd, P., Clarysse, P., Croisille, P., Nash, M. (Eds.), Biomechanics of Living Organs. (*In preprint*)
2. Miller R, Marlevi D, Zhang W, Hirschvogel M, Hadjicharalambous M, Capilnasiu A, Balmus M, Hager S, Jilberto J, **Bonini M**, Wittgenstein A, Ahmed Y, & Nordsletten D. (2021). Modeling Biomechanics in the Healthy and Diseased Heart. In J. Málek & E. Süli (Eds.), Modeling Biomaterials (pp. 141–239). Springer International Publishing.

SELECTED PRESENTATIONS

1. **Mia Bonini**, *Patient-Specific Hemodynamic Modeling for Patients with Left Ventricle Assist Device. Functional Imaging and Modeling of the Heart*. FIMH 2025. June 2025. Dallas, TX. (*podium*)
2. **Mia Bonini**, *Modeling the Hemodynamic Impact of Aortic Root Enlargements in Aortic Valve Replacement*. WCCM-PANACM. July 2024. Vancouver, Canada. (*podium*)
3. **Mia Bonini**, *Modeling the Hemodynamic Impact of Aortic Root Enlargements in Aortic Valve Replacement*. ECCOMAS. June 2024. Lisbon, Portugal. (*podium*)
4. **Mia Bonini**, *Image-Based, Whole System Hemodynamic Modeling for Valve Heart Diseases*. 22nd Computational Fluids Conference. April 2023. Cannes, France. (*podium*)
5. **Mia Bonini**, *Hemodynamic Modeling for Mitral Regurgitation*. 9th World Congress of Biomechanics. July 2022. Taipei, Taiwan. (*podium*)
6. **Mia Bonini**, *Patient Specific Hemodynamic Modeling of Transcatheter Aortic Valve Replacement*. 9th World Congress of Biomechanics. July 2022. Taipei, Taiwan. (*podium*)
7. **Mia Bonini**, *Hemodynamic Modeling for Mitral Regurgitation*. 11th European Solid Mechanics Conference. July 2022. Galway, Ireland. (*podium*)

HONORS AND AWARDS

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| 1. Functional Imaging and Modeling of the Heart Early-Career Scholarship Award, \$645 | Jun. 2025 |
| 2. Marvin Kirsh Resident Research Symposium, 1 st Place Basic Science, \$300 | Mar. 2025 |
| 3. University of Michigan Rackham Conference Travel Grant, \$1500 (x5) | 2021-2025 |
| 4. U.S. Association for Computational Mechanics Travel Award, \$775 | Jul. 2024 |
| 5. University of Michigan Rackham Graduate Student Research Grant, \$3000 | Nov. 2023 |
| 6. EUROMECH Young Researcher Award | Jul. 2022 |

LEADERSHIP AND SERVICE

- Trained and mentored 14 undergraduate and MS students in all stages of cardiovascular computational modeling, enhancing technical proficiency and research capabilities.
- Guided 3 students in preparing and presenting research at the Summer Biomechanics, Bioengineering, and Biotransport Conference, increasing their professional exposure.
- Led a student through the process of writing a research paper, resulting in a co-authored publication
- Planned and led community engagement events for the department's graduate students
- Mentored two prospective PhD students from non-traditional paths or historically disadvantaged backgrounds during the PhD application process